

Vaishnavi Sunil Pophale

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Education

Carnegie Mellon University, School of Computer Science

Master of Computational Data Science (MCDS)

Current Coursework: Machine Learning, Advanced Natural Language Processing, Interactive Data Science

Pittsburgh, PA

December 2027

Vishwakarma Institute of Information Technology (CGPA: 9.88/10 - Gold Medalist)

Bachelor of Technology in Computer Engineering | Honors in Blockchain Technology

Pune, India

June 2024

Skills

Languages and Frameworks: Python, TypeScript, SQL, FastAPI, Node.js, React.js, Model Context Protocol (MCP)

Systems and Cloud: System Design, Azure, Redis, Snowflake, Vector Databases, CI/CD, YAML

AI and ML: Natural Language Processing, Generative AI, Google Agent Development Kit (ADK), LangChain

Experience

MSCI Inc. (Morgan Stanley Capital International)

Technology Analyst | Analytics Engineering

Pune, India

Jun 2024 - Jul 2026

- Delivered a full-stack Portfolio Hedging Basket platform in cross-functional collaboration using React.js, TypeScript, and Node.js, with a configurable risk optimization engine to help traders algorithmically construct constrained hedge baskets to minimize portfolio risk.
- Architected FastAPI services processing **multi-billion-row MSCI Analytics datasets** in Snowflake; optimized data models, asynchronous processing, SQL query optimization, pagination, serialization, reducing API latency from **40s to <10s**.
- Proposed and implemented a **FastMCP**-based tool layer exposing analytics and hedging APIs to a chatbot, enabling traders to construct hedge baskets and perform risk analysis through natural language, accelerating workflows.
- Orchestrated an **AI-driven framework upgrade pipeline** enabling automated version discovery, repository-wide dependency analysis, compatibility checks, code upgrades, and build validation across centralized modules.
- Designed a multi-tenant Redis caching layer for **low latency access to 300K+ portfolios per tenant**, with automated cache lifecycle management and data isolation, reducing Snowflake queries, API calls, and compute overhead.
- Eliminated manual credential provisioning and per-client, per-environment secret storage in Azure Key Vault by engineering a fast, real-time credential retrieval system with secret-expiry handling and request coalescing, preventing access issues.

Technology and Data Intern

Jan 2024 - Jun 2024

- Initiated and led a **zero-downtime** Azure configuration framework integration for the Portfolio Insights platform supporting **200+ institutional clients**, using CI/CD pipelines and YAML configuration, enabling runtime, environment-aware, and client-specific configuration updates without code deployments or service restarts.

Indian Space Research Organisation (ISRO) - Vikram Sarabhai Space Centre

Project Intern

Trivandrum, India

Oct 2023 - Nov 2023

- Automated complex load-case generation for aerospace and mechanical structural analysis software using pybind11, eliminating repetitive manual setup and enabling engineers to run more scenarios faster, with fewer errors.

Projects

- Financial Data MCP Server:** Developed a FastAPI-MCP financial server over the Yahoo Finance dataset, exposing schema-preserving MCP tools through asynchronous FastAPI endpoints and Pydantic schemas for agentic LLM workflows.
- Multi-Agent Orchestration System:** Built a multi-agent AI system using Google Agent Development Kit, orchestrating agents via task delegation using the Agent-as-a-Tool pattern with Gemini-backed reasoning and custom tool integration.
- Confluence RAG Knowledge Assistant:** Engineered a LangChain-powered RAG pipeline ingesting Confluence documentation into ChromaDB with Azure OpenAI, enabling context-aware question answering and semantic retrieval.

Publications

- Question and Assessment Generator: Deep Learning Approach for Assessment Creation - *IEEE Xplore, ICWITE 2024*
- State of the Art Analysis of Word Sense Disambiguation - *Springer, ICBDS 2023*
- IPFS-Blockchain Technology for Lease Agreement and Rental System in Real Estate - *IEEE Xplore, ICBDS 2023*
Best Research Paper Award
- A Survey and Implementation of Machine Learning Algorithms for Customer Churn Prediction - *IJRITCC 2023*